

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 6: Solving Problems Based on Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Problems on speed, distance and time

1. A couple drive 350 km to a hill station. Heavy rain reduces their average speed by 20 km/h, and the journey takes 2 hours longer. Find (a) the original speed of the car (b) the time taken at the reduced speed. **ICSE 2025, 4 marks**

SOLUTION

(a) Let the original speed be x km/h. $\frac{350}{x-20} - \frac{350}{x} = 2$
 $350x - 350(x-20) = 2x(x-20)$, so $7000 = 2x^2 - 40x$ and $x^2 - 20x - 3500 = 0$
 $(x-70)(x+50) = 0$, so $x = 70$ (speed cannot be negative).
 (b) Reduced speed = 50 km/h, so the time is $\frac{350}{50} = 7$ hours.

Answer: (a) 70 km/h (b) 7 hours

2. A man travels 100 km at a uniform speed of x km/h. At a speed 5 km/h faster, the journey would have taken 1 hour less. Find x . **ICSE 2023, 3 marks**

SOLUTION

$\frac{100}{x} - \frac{100}{x+5} = 1$
 $100(x+5) - 100x = x(x+5)$, so $x^2 + 5x - 500 = 0$
 $(x+25)(x-20) = 0$, so $x = 20$ (speed cannot be negative).

Answer: 20 km/h

2 Problems on numbers

3. Two natural numbers differ by 7 and their product is 450. Find the numbers. **ICSE 2020, 3 marks**

SOLUTION

Let the smaller number be x . Then $x(x+7) = 450$, so $x^2 + 7x - 450 = 0$.
 $(x+25)(x-18) = 0$, so $x = 18$ (a natural number cannot be negative).

Answer: 18 and 25

4. Two consecutive multiples of 3 are natural numbers with a product of 810. Find them.

ICSE 2019, 3 marks

SOLUTION

Let the numbers be $3x$ and $3x + 3$. Then $3x(3x + 3) = 810$, so $x(x + 1) = 90$.

$x^2 + x - 90 = 0$, so $(x + 10)(x - 9) = 0$ and $x = 9$.

Answer: 27 and 30

3 Problems on cost and on work

5. ₹7500 is divided equally among some children. With 20 fewer children, each would have received ₹100 more. Find the original number of children. ICSE 2018, 3 marks

SOLUTION

Let there be x children. $\frac{7500}{x - 20} - \frac{7500}{x} = 100$

$7500 \times 20 = 100x(x - 20)$, so $x^2 - 20x - 1500 = 0$

$(x - 50)(x + 30) = 0$, so $x = 50$.

Answer: 50 children

4 Problems on ages

6. The ages of two brothers add up to 47 years, and the product of their ages is 550. Find their ages. ICSE 2017, 4 marks

ICSE 2017, 4 marks

SOLUTION

Let the elder brother be x years old. Then the younger is $(47 - x)$ years old.

$x(47 - x) = 550$, so $x^2 - 47x + 550 = 0$

$(x - 25)(x - 22) = 0$, so $x = 25$ or 22 . The elder is 25.

Answer: 25 years and 22 years

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